

Multiple Choice

1. **Answer:** D

Options: A) 0.75 m/sec; B) 3.5 m/sec; C) 0.5 m/sec; D) 1.5 m/sec; E) 6 m/sec

Note: Correct option: D - 1.5 m/sec

2. **Answer:** A

Options: A) 2.25; B) 2.0; C) 2.5; D) 1.41; E) 0.5

Note: Correct option: A - 2.25

3. **Answer:** A

Options: A) 8.5×10^{11} ; B) 7.1×10^{11} ; C) 7.5×10^{11} ; D) 9.2×10^{11} ; E) 3.8×10^{11}

Note: Correct option: A - 8.5×10^{11}

4. **Answer:** A

Options: A) decreases slightly; B) increases but does not reach double; C) remains the same; D) halves; E) quadruples

Note: Correct option: A - decreases slightly

5. **Answer:** C

Options: A) dependent on the speed of the wind; B) greater than 0.8 N; C) 0.8 N; D) dependent on the temperature of the air; E) dependent on the orientation of its body

Note: Correct option: C - 0.8 N

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.85'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '5.55'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '600,000 dynes'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1100 MeV'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The force of the scale on the man'.

Long Form Response

11. **Answer:** 2/5. Explain why this is the correct answer and provide supporting evidence.

Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1 ft. Explain why this is the correct answer and provide supporting evidence.

Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.